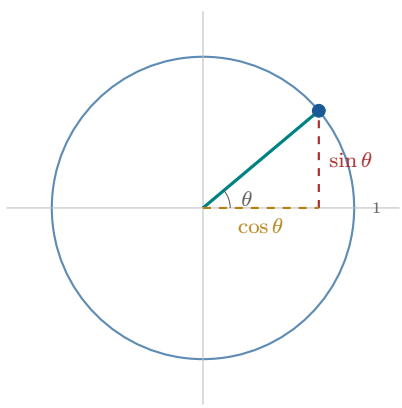
Constraint and Transformation

An Integrated Introduction to

Algebra, Geometry, Trigonometry, and Chemical Stoichiometry



$\sin^2\theta + \cos^2\theta = 1$

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Constraint and Transformation
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Notation and Symbols

The following symbols are used consistently throughout the text. A change of meaning is always announced explicitly.

Symbol	Meaning
$\mathbb{R}$	real number line
$\mathbb{N}$	natural numbers $\{0, 1, 2, \dots\}$
x, y, z	generic real variables
a, b, c	constants or geometric side lengths
f, g, h	function names
θ	angle (radians unless noted)
r	radius of a circle or sphere
n	amount of substance (moles, mol)
M	molar mass (g/mol)
m	mass (grams or kilograms)
L	luminosity of a star (watts)
I	radiation intensity ($\mathrm{W/m^2}$)
d	distance
A	area
V	volume
Φ	scalar field (RSVP plenum density)
$\mathbf{v}$	vector flow field
S	entropy density field
γ	stoichiometric coefficient vector
N	element–species incidence matrix
$\mathcal{J}$	variational functional
$\preceq$	causal priority order on events

Preface

This book is built on one conviction: algebra, geometry, trigonometry, and chemistry are not separate disciplines accidentally packaged together by tradition. They are four dialects of the same language — the language of *constraint and transformation*.

Every equation encountered here is a claim about balance. Algebra encodes symbolic balance among quantities. Geometry encodes spatial balance among dimensions. Trigonometry encodes rotational and cyclic balance. Stoichiometry encodes material balance in chemical reactions. To understand any one of these is to recognize a pattern that recurs across all the others: a system remains coherent because its constitutive relations must simultaneously be satisfied.

This orientation reflects a structural feature of scientific reasoning. Quantitative systems are defined by relations that must remain internally consistent as quantities change. A child discovering that a toy machine follows a hidden rule, an astrophysicist reading the composition of a star from its spectrum, a political theorist analyzing how collective power emerges from coordination, a novelist constructing an ecologically stable imaginary world — each is solving, in different vocabulary, the same problem: *what constraints govern this system, and what transformations preserve them?*

Intellectual influences. The book's perspective has been shaped by several intellectual traditions. The developmental psychology of Gopnik [2] shows how causal inference in children is structurally isomorphic with algebraic constraint satisfaction. The astrophysics of Payne-Gaposchkin [8] demonstrates that geometric and spectroscopic reasoning can reveal chemical composition at stellar distances. The political philosophy of Arendt [1] offers a model in which social systems are organized by relational structure rather than by fixed hierarchies. The complexity science of Juarrero [5] formalizes how constraints propagate through dynamical systems, while the speculative fiction of Guin [3] explores the societal consequences of such constraint structures with rare concreteness. The visual mathematical approach of Needham [6, 7] informs the book's emphasis on diagrammatic reasoning.

How to read this book. Each chapter follows a three-step rhythm: *observation* (a phenomenon in plain language), *formalization* (the mathematical statement), and *extension* (the same idea in a new domain). The main text is self-contained for an introductory reader. Each chapter ends with three tiers of exercises: **Practice** (computation), **Exercise** (conceptual reasoning), and **Challenge** (open-ended exploration). The appendices develop the deeper formal architecture — field theory, category theory, sheaf semantics, variational principles, and recursive simulation — for readers who wish to see how elementary mathematics extends into contemporary research.

Prerequisites. Arithmetic: fractions, decimals, percentages, signed numbers. No prior algebra, geometry, or chemistry is assumed.

Chapter 1

Mathematics as Structured Relation

1.1 The Central Thesis

Scientific descriptions can be viewed as networks of relationships rather than lists of isolated facts. This book proposes that the four subjects it covers — algebra, geometry, trigonometry, and chemistry — are four languages for a single idea: a system is defined not merely by its components but by the *relations* those components must satisfy.

Definition Constraint

A **constraint** is any condition restricting the possible values of one or more quantities. Formally, given variables $x_1, \ldots, x_n$, a constraint is a relation

$$F(x_1, \ldots, x_n) = 0.$$

The **solution set** is $\mathcal{C} = \{(x_1, \ldots, x_n) \in \mathbb{R}^n \mid F(x_1, \ldots, x_n) = 0\}$.

Guiding Principle

Every equation in this book is a claim that two descriptions of the same situation must agree. Solving the equation means finding the values that live inside the solution set $\mathcal{C}$.

1.2 Four Languages, One Pattern

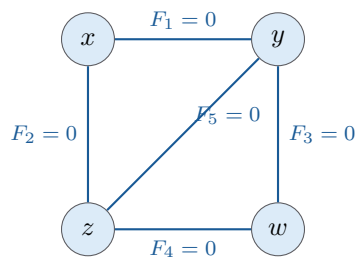
The table below introduces the central motif. Each row names a type of constraint, the domain where it appears, and a representative equation.

Domain	Constraint type	Canonical equation
Algebra	symbolic / numerical	$2x + 3 = 11$
Geometry	spatial / dimensional	$a^2 + b^2 = c^2$
Trigonometry	cyclic / angular	$\sin^2\theta + \cos^2\theta = 1$
Stoichiometry	material / atomic	$2H_2 + O_2 \to 2H_2O$

In every case the equation is a *conservation statement*: it asserts that some quantity — a sum, a squared length, a trigonometric magnitude, or an atom count — is preserved under transformation.

1.3 Constraint Networks

A single equation imposes one constraint. Real systems impose many constraints simultaneously. Representing them as a network reveals their mutual dependencies.



Each node represents a quantity; each edge represents a constraint relating the two adjacent quantities. A consistent system is one in which all constraints can be satisfied simultaneously.

Deeper Structure

The general form of a linear constraint system is $A\mathbf{x} = \mathbf{b}$, where A is an $m \times n$ matrix. Appendix A develops this idea formally, including its connections to the RSVP field framework and Spherepop event-history calculus.

1.4 The Book's Arc

The book progresses from local to global. Part I develops algebraic constraints on individual quantities. Part II extends this to spatial relationships. Part III describes cyclic and rotational constraints. Part IV applies all of this to chemical systems, which are constraint networks over atomic species. Part V synthesizes the four domains into a unified picture of structured systems.

Exercises

Exercise 1.1. Write down three everyday relationships and express each as an equation $F(x, y) = 0$. What does the solution set mean in each case?

Challenge 1.2. Draw a constraint network with five quantities and at least six constraints. Choose a real system (a recipe, a budget, a simple machine) as your model. Explain why some constraints are necessary and others may be redundant.

Part I

Algebra: Relations and Constraint Systems

Chapter 2

Numbers, Variables, and Representation

2.1 Numbers as Quantities

A **number** is a symbolic stand-in for a measurable quantity: a count, a length, a temperature. Numbers obey arithmetic: addition, subtraction, multiplication, division.

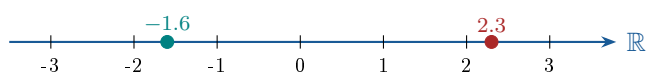
Definition Variable

A **variable** is a letter that stands for an unknown or varying quantity. An **expression** is a combination of numbers and variables using arithmetic operations. An expression does not assert equality; it simply describes a quantity.

The expression $3x + 5$ has a specific numerical value for each specific value of x. It is not an equation and does not claim anything — it is simply a description.

2.2 The Real Number Line

Every real number corresponds to a unique point on an infinite line. Distance from the origin measures magnitude; direction indicates sign.



2.3 Proportionality: The Simplest Relation

Theorem 2.1 (Direct Proportionality)**.** *If two quantities x and y satisfy $y = kx$ for a constant $k \neq 0$, we say y is* ***directly proportional*** *to x with proportionality constant k. The graph of this relation is a straight line through the origin.*

Worked Example

A car travels at a constant speed of 60 km/h. Distance d (km) and time t (h) satisfy

$$d = 60\,t.$$

Checking units: km = (km/h) · h. ✓
The slope $k = 60$ is the rate of change: each additional hour yields 60 more kilometres. For $t = 2.5$ h, $d = 150$ km.

Historical Note Alison Gopnik and Causal Inference

In her research on how children understand the world, Alison Gopnik showed that even toddlers spontaneously construct algebraic-style rules from observation [2]. A child who notices that a machine lights up whenever two specific blocks are inserted together is, in effect, discovering a proportional or Boolean rule governing the system. Elementary algebra makes this intuition precise.

Exercises

Practice 2.1. A recipe uses 3 cups of flour per 2 cups of sugar. Write an equation relating flour f to sugar s. How much sugar is needed for 9 cups of flour?

Exercise 2.2. A toy machine activates according to $A = 3B + 2$, where B is the number of blue blocks inserted. If the observed activation value is $A = 14$, determine B and interpret the result.

Challenge 2.3. Collect five proportional relationships from daily life (speed and distance, ingredients in a recipe, etc.) and estimate each proportionality constant. Which relationships are only approximately proportional? What causes the deviation?

Chapter 3

Equations and Balance

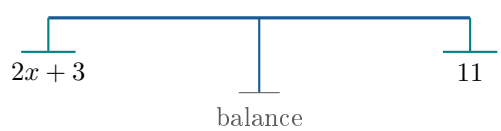
3.1 Equations as Statements of Equality

Definition Equation and Solution

An **equation** is a statement $A = B$ where A and B are expressions. A **solution** is any assignment of values to the variables that makes the statement true.

An equation is therefore a constraint: it asserts that exactly one value (or set of values) satisfies the balance condition.

3.1.1 The Balance Model



Whatever operation is applied to one side of the equation must be applied identically to the other — the scale must remain balanced.

3.2 Solving Linear Equations

Theorem 3.1 (Isolation Principle)**.** *A linear equation $ax + b = c$ (with $a \neq 0$) has the unique solution*

$$x = \frac{c - b}{a}.$$

Proof Sketch. Subtract b from both sides: $ax = c - b$. Divide both sides by $a \neq 0$. □

Worked Example Balancing a simple equation

Solve $2x + 3 = 11$.

$$\begin{aligned} 2x + 3 &= 11 && \text{original constraint} \\ 2x &= 8 && \text{subtract } 3 \\ x &= 4 && \text{divide by } 2 \end{aligned}$$

Verification: $2(4) + 3 = 11$. ✓

Worked Example Gopnik: Hypothesis testing as equation solving

A child hypothesises that a toy machine follows the rule $A = 3G + Y$, where G and Y are the numbers of green and yellow blocks. She observes $A = 8$ and inserts $Y = 2$. Solving for G:

$$3G + 2 = 8 \implies 3G = 6 \implies G = 2.$$

3.3 Dimensional Analysis

Guiding Principle

Every valid equation involving physical quantities must be **dimensionally consistent**: the units on both sides must be identical.

Worked Example Converting units

Convert 72 km/h to m/s:

$$72\ \frac{\text{km}}{\text{h}} \times \frac{1000\ \text{m}}{1\ \text{km}} \times \frac{1\ \text{h}}{3600\ \text{s}} = 20\ \frac{\text{m}}{\text{s}}.$$

Exercises

Practice 3.1. Solve each equation.
(a) $5x - 7 = 18$ (b) $\frac{x}{3} + 4 = 9$ (c) $-2x + 10 = 0$ (d) $3(x - 2) = 15$.

Exercise 3.2. A reaction produces 0.5 mol of gas per minute. Write the constraint equation and determine how many moles are produced in 2.5 hours. Include units at every step.

Challenge 3.3. Arendt [1] argues that political power P emerges from coordination, not mere numbers. Model $P = cn$ where c is coordination strength and n is the number of participants. If $c = 0.2$ initially and the group triples its coordination to $c = 0.6$, by what factor does P change? What does this say about collective action?

Chapter 4

Proportions and Scaling

4.1 Ratios and Proportions

A **ratio** compares two quantities. A **proportion** asserts that two ratios are equal:

$$\frac{a}{b} = \frac{c}{d}.$$

Worked Example Stellar mass ratios

In the hydrogen fusion reaction $4\text{H} \to \text{He}$, the mass ratio of hydrogen consumed to helium produced is approximately $4:1$ (by nucleon count). If a stellar region contains 10^{12} hydrogen nuclei per second, the production rate of helium nuclei satisfies

$$\frac{n_{\text{He}}}{1} = \frac{n_{\text{H}}}{4} \implies n_{\text{He}} = \frac{10^{12}}{4} = 2.5 \times 10^{11} \text{ nuclei/s}.$$

Historical Note Cecilia Payne-Gaposchkin

Cecilia Payne-Gaposchkin (1900–1979) discovered in her 1925 doctoral dissertation that stars are composed primarily of hydrogen and helium — a result initially rejected by the astronomical establishment but later recognised as one of the most important discoveries in astrophysics [8]. Her analysis relied on proportional reasoning connecting spectral line intensities to elemental abundances.

4.2 Scaling Laws

Theorem 4.1 (Power Scaling)**.** *If quantity y scales as the p-th power of quantity x, then $y = kx^p$. When x is multiplied by a factor λ, y is multiplied by λ^p.*

Worked Example Le Guin: island resource capacity

In a circular island settlement of radius $r = 2$ km, the total area is $A = \pi r^2 = 4\pi \approx 12.6$ km^2. If agricultural land occupies half the island and each km^2 supports 500 people:

$$\text{capacity} = 500 \times \frac{4\pi}{2} \times 100 = 500 \times 200\pi \approx 314{,}159 \text{ people.}$$

Doubling the radius to $r = 4$ km quadruples the area and thus quadruples the capacity. Scaling is not linear [3].

Exercises

Practice 4.1. A car uses 8 L per 100 km. How far can it travel on 22 L?

Exercise 4.2. A solution is prepared at 5 g of salt per 200 mL. How much salt is needed for 500 mL? Write the proportion and solve.

Challenge 4.3. Le Guin's utopian novel *The Dispossessed* [3] depicts a society on an arid moon that must balance population against scarce resources. Suppose the moon's habitable area is A km^2, each km^2 supports at most p people, and the society aims to keep population at 80% of capacity. Express the target population as a function of A and p. How does the target change if a drought reduces A by 15%?

Chapter 5

Systems of Equations

5.1 Multiple Constraints, Multiple Unknowns

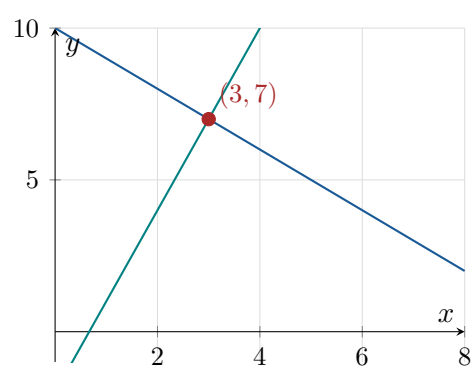
When a situation involves two or more unknowns, a single equation constrains but does not determine. A system of n equations in n unknowns typically yields a unique solution.

Definition System of Equations

A **system of m equations in n unknowns** is a collection $\{F_1 = 0, \ldots, F_m = 0\}$ that must all hold simultaneously. The **solution** is any tuple satisfying every equation at once.

5.2 Geometric Interpretation

Each linear equation $ax + by = c$ defines a **line** in the xy-plane. A system of two such equations corresponds to two lines; the solution is their intersection.



Worked Example Juarrero: constraint propagation through a system

Juarrero [5] emphasises that constraints propagate through systems: a change in one relation

alters the whole. Consider a simple model:

$$\begin{cases} F = 2L \\ E = F + L, \end{cases}$$

with $L = 100$. Then $F = 200$, $E = 300$. If a drought reduces L to 80, then $F = 160$ and $E = 240$. The constraint cascade propagates through the entire system.

Worked Example Solving by elimination

Solve $\{x + y = 10,\ 3x - y = 2\}$.
Adding the two equations: $4x = 12$, so $x = 3$. Then $3 + y = 10$, so $y = 7$.
Verification: $3 + 7 = 10$ ✓; $9 - 7 = 2$ ✓.

Exercises

Practice 5.1. Solve the system $\{2x + 3y = 12,\ x - y = 1\}$ by both substitution and elimination.

Exercise 5.2. Interpret geometrically: (a) a system with no solution; (b) a system with infinitely many solutions. What does each case mean for the constraint network?

Challenge 5.3. A chemist mixes Solution A (20% acid) and Solution B (50% acid) to produce 300 mL of 35% acid. Set up and solve the system. Connect this to Juarrero's idea of constraints propagating through a mixture network.

Chapter 6

Matrices and Linear Transformations

6.1 Systems in Matrix Form

A system $\{a_1x + b_1y = c_1,\ a_2x + b_2y = c_2\}$ may be written as

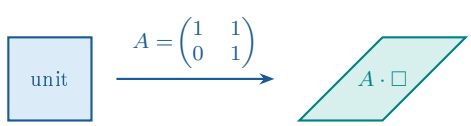
$$A\mathbf{x} = \mathbf{b}, \qquad A = \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}.$$

Definition Matrix

An $m \times n$ **matrix** is a rectangular array of mn numbers arranged in m rows and n columns. It encodes a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ by $T(\mathbf{x}) = A\mathbf{x}$.

6.2 Matrices as Transformations of Space

A 2×2 matrix A transforms every point (x, y) in the plane. The unit square is sent to a parallelogram.



6.3 The Stoichiometric Matrix

Chemical reactions are encoded by a stoichiometric matrix N whose rows index elements and whose columns index species.

Worked Example Water formation as matrix equation

For $2H_2 + O_2 \to 2H_2O$, writing $\gamma = (-2, -1, 2)^\top$ (negative for reactants):

$$N = \begin{pmatrix} 2 & 0 & 2 \\ 0 & 2 & 1 \end{pmatrix}, \qquad N\gamma = \begin{pmatrix} 2(-2) + 0(-1) + 2(2) \\ 0(-2) + 2(-1) + 1(2) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Conservation is encoded by $N\gamma = \mathbf{0}$.

Deeper Structure

Appendix D develops stoichiometry as conservation algebra in full generality, including the reaction rate equation $\dot{c} = Sr(c)$.

Exercises

Practice 6.1. Multiply $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 4 \\ -1 \end{pmatrix}$.

Exercise 6.2. Write the system $\{x+2y = 7,\ 3x-y = 5\}$ as $A\mathbf{x} = \mathbf{b}$ and solve using Cramer's rule or row reduction.

Challenge 6.3. Model Arendt's power equation $P = C\mathbf{n}$ as a matrix transformation, where $\mathbf{n}$ is a vector of participants across three groups and C is a 1×3 coordination matrix. What does the rank of C represent?

Part II

Geometry: Spatial Structure and Invariance

Chapter 7

Points, Lines, and Distance

7.1 The Coordinate Plane

René Descartes' insight was that geometric objects can be described algebraically. A point is an ordered pair (x, y); a line is the solution set of a linear equation; a circle is the solution set of a quadratic relation.

Theorem 7.1 (Distance Formula)**.** *The distance between points (x_1, y_1) and (x_2, y_2) in the plane is*

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

This formula is a consequence of the Pythagorean theorem, which we derive in the next chapter.

7.2 Angles and Their Measure

Definition Radian

One **radian** is the angle subtended at the centre of a circle by an arc equal in length to the radius. Since the full circumference is $2\pi r$, a full rotation equals 2π radians.

Conversion: $\theta_{\text{rad}} = \frac{\pi}{180}\theta_{\text{deg}}$.

Exercises

Practice 7.1. Convert: (a) 60 to radians; (b) $\frac{3\pi}{4}$ to degrees.

Exercise 7.2. Show algebraically that the midpoint of the segment from (x_1, y_1) to (x_2, y_2) is $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$.

Chapter 8

Triangles and the Pythagorean Relation

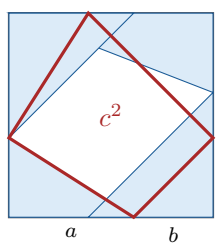
8.1 Right Triangles

Theorem 8.1 (Pythagorean Theorem)**.** *In any right triangle with legs a, b and hypotenuse c,*

$$a^2 + b^2 = c^2.$$

The length of the hypotenuse is completely determined by the two legs.

Proof Sketch. Place four congruent right triangles inside a square of side $a+b$. The uncovered central region is a square of side c.



Area of outer square: $(a+b)^2 = a^2 + 2ab + b^2$. Four triangles: $4 \cdot \frac{1}{2}ab = 2ab$. Inner square: $c^2 = (a+b)^2 - 2ab = a^2 + b^2$. □

8.2 Applications

Worked Example Payne-Gaposchkin: radiation and distance

Light from a star of luminosity L (W) spreads uniformly over a sphere of radius r. Surface area $A = 4\pi r^2$ (m^2), so intensity is

$$I = \frac{L}{4\pi r^2} \text{ W/m}^2.$$

For the Sun: $L = 3.8 \times 10^{26}$ W at $r = 1.5 \times 10^{11}$ m:

$$4\pi r^2 \approx 2.83 \times 10^{23}\ \mathrm{m}^2, \qquad I \approx 1340\ \mathrm{W/m}^2.$$

The inverse-square law is a constraint on spatial geometry.

Exercises

Practice 8.1. Find the diagonal of a 7×24 cm rectangle.

Exercise 8.2. Two ships leave the same port. One travels 40 km due north; the other 30 km due east. How far apart are they? Draw and label the constraint diagram.

Challenge 8.3. Star A has luminosity $2L$ and is at distance $3d$. Star B has luminosity L and is at distance d. Which appears brighter to an observer, and by what factor? Generalise: express the apparent brightness ratio as a function of the luminosity and distance ratios.

Chapter 9

Similarity and Scaling in Geometry

9.1 Similar Figures

Definition Similarity

Two figures are **similar** (written $\sim$) if one can be obtained from the other by scaling, rotation, or reflection. Corresponding angles are equal; corresponding lengths are proportional with scale factor k.

Theorem 9.1 (Scaling Laws)**.** *If two similar figures have linear scale factor k, then areas scale as k^2 and volumes scale as k^3.*

Proof Sketch. Area involves two length dimensions; volume involves three. Each scales as a power of k equal to the number of dimensions. □

Exercises

Practice 9.1. Two similar triangles have scale factor $3:5$. If the smaller has area $27\ \text{cm}^2$, find the area of the larger.

Exercise 9.2. A spherical star with radius r has surface area $4\pi r^2$. If the star expands to radius $2r$, by what factor does its surface area increase? Its volume?

Chapter 10

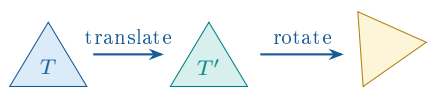
Transformations of the Plane

10.1 Isometries: Transformations Preserving Distance

A **transformation** is a function mapping the plane to itself. An **isometry** preserves distances: $d(T(P), T(Q)) = d(P, Q)$ for all points P, Q.

The three basic isometries are:

- **Translation** by vector $\mathbf{v}$: $T(P) = P + \mathbf{v}$.
- **Rotation** by angle θ around origin: encoded by $R(\theta)$.
- **Reflection** across a line ℓ.



Guiding Principle

Every isometry preserves shape and size. It is a constraint-preserving transformation: the distance relation $d(P, Q) = c$ becomes $d(T(P), T(Q)) = c$.

Historical Note Hannah Arendt and Invariance

Arendt's analysis of political action distinguishes between acts that alter the social space and those that maintain it. An isometry is the spatial analogue of a norm-preserving act: the shape of relations is maintained even as positions change. The rotation matrix $R(\theta)$ is the canonical mathematical example.

Exercises

Exercise 10.1. Show that the composition of two rotations around the origin is another rotation. What is the resulting angle?

Challenge 10.2. A rotation by θ is represented by the matrix $R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$. Compute $R(\theta)^\top R(\theta)$ and explain why this confirms that $R(\theta)$ is an isometry.

Chapter 11

Area, Volume, and Spatial Distribution

11.1 Area Formulas as Algebraic Constraints

Shape	Formula	Constraint expressed
Rectangle	$A = lw$	product of perpendicular dimensions
Triangle	$A = \frac{1}{2}bh$	half the enclosing rectangle
Circle	$A = \pi r^2$	area-radius relation
Sphere	$A = 4\pi r^2$	surface of constant curvature
Cylinder	$V = \pi r^2 h$	circular cross-section times height

Worked Example Le Guin: sustainable population

A circular settlement of radius $r = 3$ km devotes 60% of its area to crops. Each km^2 supports 400 people.

$$A = \pi(3)^2 = 9\pi \text{ km}^2, \qquad A_c = 0.6 \times 9\pi = 5.4\pi \text{ km}^2.$$

Population capacity: $400 \times 5.4\pi \times 100 \approx 679{,}000$. (Factor of 100 converts km^2 to hectares.) Le Guin's ecological societies are governed by exactly such spatial constraints [3].

Exercises

Practice 11.1. A circle has area 50π cm^2. Find its circumference.

Exercise 11.2. A cylindrical grain silo has radius 2 m and height 6 m. Find its volume and total surface area. Interpret each quantity in the context of grain storage.

Part III

Trigonometry: Cycles and Rotational Systems

Chapter 12

Angles and Circular Geometry

12.1 The Circle as Generator of Cycles

A circle of radius r centred at the origin is the solution set of $x^2 + y^2 = r^2$. As a point moves around the circle, its x- and y-coordinates each trace out a periodic oscillation. Trigonometry is the mathematics of this periodicity.

Historical Note Cyclic patterns in astrophysics

Cecilia Payne-Gaposchkin also studied Cepheid variable stars, whose brightness oscillates with remarkable regularity. The period of oscillation encodes the star's intrinsic luminosity — a fact used to measure cosmic distances. Trigonometric modelling of such cycles was central to her work [8].

Exercises

Practice 12.1. State the arc length formula $s = r\theta$ (where θ is in radians). Find the arc length subtended by an angle of $\pi/3$ on a circle of radius 5 cm.

Chapter 13

Sine, Cosine, and the Unit Circle

13.1 The Fundamental Definitions

Definition Sine and Cosine

For a point on the unit circle at angle θ from the positive x-axis,

$$\cos\theta = x\text{-coordinate}, \qquad \sin\theta = y\text{-coordinate}.$$

Equivalently, in a right triangle with adjacent side a, opposite side o, and hypotenuse h:

$$\cos\theta = \frac{a}{h}, \qquad \sin\theta = \frac{o}{h}, \qquad \tan\theta = \frac{o}{a}.$$

Theorem 13.1 (Pythagorean Identity)**.** *For all θ,*

$$\sin^2\theta + \cos^2\theta = 1.$$

The squared components of a unit vector always sum to one.

Proof Sketch. A point $(\cos\theta, \sin\theta)$ lies on the unit circle by definition. The circle's equation $x^2 + y^2 = 1$ gives the identity directly. □

13.2 Exact Values

θ	radians	$\sin\theta$	$\cos\theta$	$\tan\theta$
0	0	0	1	0
30	$\pi/6$	$1/2$	$\sqrt{3}/2$	$1/\sqrt{3}$
45	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1
60	$\pi/3$	$\sqrt{3}/2$	$1/2$	$\sqrt{3}$
90	$\pi/2$	1	0	undef

Exercises

Practice 13.1. Without a calculator, compute $\sin(120)$ and $\cos(135)$.

Exercise 13.2. Prove that $\sin(90 - \theta) = \cos\theta$ using the unit circle definitions.

Challenge 13.3. The rotation matrix $R(\theta)$ preserves the norm of every vector. Express this statement as an identity involving $\sin\theta$ and $\cos\theta$, and relate it to the Pythagorean identity.

Chapter 14

Trigonometric Functions and Graphs

14.1 Periodic Functions

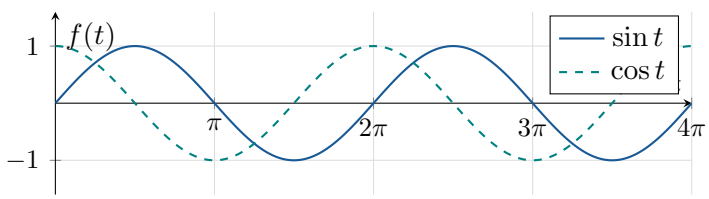
Definition Period

A function f is **periodic** with period $T > 0$ if $f(t+T) = f(t)$ for all t. The sine and cosine functions have period 2π.

The general sinusoidal function

$$f(t) = A\sin(\omega t + \phi) + D$$

has **amplitude** A, **angular frequency** ω, **phase** ϕ, and **vertical shift** D. The period is $T = 2\pi/\omega$.



14.2 Variable Stars: Trigonometric Modelling

Worked Example Payne-Gaposchkin: stellar brightness cycles

A Cepheid variable star's brightness is modelled by

$$B(t) = 10 + 2\sin\left(\frac{2\pi}{5}t\right),$$

with amplitude 2 and equilibrium brightness = 10.
At $t = 2$ days:

$$B(2) = 10 + 2\sin\left(\frac{4\pi}{5}\right) = 10 + 2\sin(144) \approx 10 + 2(0.588) = 11.18.$$

This model was used by Payne-Gaposchkin to link oscillation period to distance measurement.

Exercises

Practice 14.1. Identify the amplitude, period, and phase of $f(t) = 3\sin(2t - \pi/4)$.

Exercise 14.2. Gopnik [2] notes that children alternate between exploration and exploitation in a roughly periodic pattern. Model the exploration intensity as $E(t) = 0.5 + 0.5\sin(\omega t)$. When is exploration at maximum? When is it at minimum? What is the period if the cycle is one week?

Chapter 15

Modeling Cyclic Phenomena

15.1 Harmonic Motion

Simple harmonic motion arises whenever a restoring force is proportional to displacement. The governing equation is

$$\frac{d^2x}{dt^2} + \omega^2 x = 0,$$

with general solution $x(t) = A\cos(\omega t) + B\sin(\omega t)$.

Worked Example Wave interference

Two waves $f_1 = A\sin(\omega t)$ and $f_2 = A\sin(\omega t + \delta)$ combine to

$$f_1 + f_2 = 2A\cos\left(\frac{\delta}{2}\right)\sin\left(\omega t + \frac{\delta}{2}\right).$$

When $\delta = \pi$, amplitude vanishes (destructive interference). The constraint $\delta = \pi$ is exactly the condition for cancellation.

Exercises

Exercise 15.1. Two sides of a triangle are 7 and 9; the included angle is 40. Use the Law of Cosines to find the third side. Show all unit checks.

Challenge 15.2. Juarrero [5] argues that organisms maintain coherent behaviour through nested cycles of constraint. Model two interacting oscillators $x(t) = \sin(t)$, $y(t) = \sin(t + \phi)$. For what value of ϕ do they cancel? For what value do they reinforce? Discuss the biological analogy.

Chapter 16

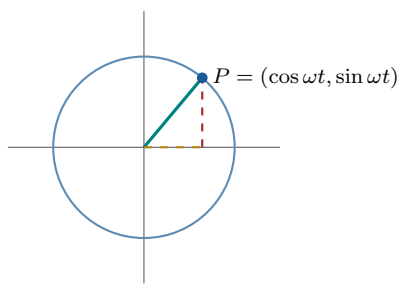
Rotations and Harmonic Motion

16.1 Circular Motion and Projections

A point moving at constant angular velocity ω on a circle of radius R has position

$$(x, y) = (R\cos(\omega t), R\sin(\omega t)).$$

The x- and y-projections are independent sinusoidal oscillations, which is why circular motion and wave motion share the same mathematics.



Exercises

Exercise 16.1. Show that if a point traces the unit circle at constant speed, its x- and y-velocities are $-\sin(\omega t)$ and $\cos(\omega t)$ respectively. Verify using the Pythagorean identity that the speed is constant.

Part IV

Stoichiometry: Conservation and Transformation in Chemistry

Chapter 17

Atoms, Molecules, and Chemical Formulas

17.1 The Atomic Hypothesis and Conservation

In ordinary chemical reactions, atoms are neither created nor destroyed — they are rearranged. This is the **law of conservation of mass**, the chemical analogue of every balance principle studied so far.

Definition Mole and Molar Mass

The **mole** (mol) is the SI unit of chemical amount: one mole contains $N_A = 6.022 \times 10^{23}$ particles (Avogadro's number). The **molar mass** M (g/mol) is the mass of one mole of a substance. The relationship

$$n = \frac{m}{M}$$

converts between mass and amount of substance.

Historical Note Payne-Gaposchkin: hydrogen and helium in stars

Payne-Gaposchkin's 1925 discovery that stars are overwhelmingly hydrogen and helium was resisted because it contradicted the then-prevailing view. Her argument was essentially stoichiometric: matching observed spectral absorption strengths to theoretical abundances. The molar proportions of H to He in a star constrain its entire energy budget [8].

Worked Example Moles of water

How many moles are present in 36 g of H_2O?

$$M(H_2O) = 2(1.008) + 16.00 = 18.016\,\text{g/mol}. \qquad n = \frac{36}{18.016} \approx 2.0\,\text{mol}.$$

Exercises

Practice 17.1. Find the molar mass of (a) NaCl; (b) CO_2; (c) $C_6H_{12}O_6$.

Exercise 17.2. How many molecules are in 44 g of CO_2? Express in scientific notation.

Challenge 17.3. Payne-Gaposchkin computed that the Sun's atmosphere is approximately 92% hydrogen by number of atoms. If the Sun's atmosphere contains N total atoms, write an expression for the number of hydrogen atoms and the number of helium atoms (assuming the remaining 8% is helium). What mole ratio does this represent?

Chapter 18

Balancing Chemical Equations

18.1 Conservation as a Linear System

Theorem 18.1 (Stoichiometric Conservation)**.** *A chemical equation is balanced if and only if, for each element, the total atom count on the left side equals the total atom count on the right side. Equivalently, the stoichiometric coefficient vector γ satisfies $N\gamma = \mathbf{0}$, where N is the element–species incidence matrix.*

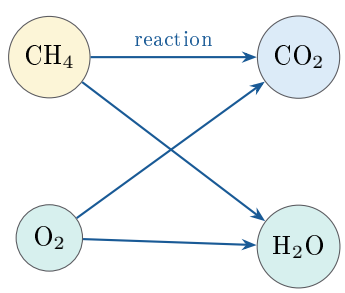
Worked Example Combustion of ethane

Balance $C_2H_6 + O_2 \to CO_2 + H_2O$.
Assign coefficients a, b, c, d and write conservation equations:

$$\begin{aligned} \mathrm{C}: &\quad 2a = c \\ \mathrm{H}: &\quad 6a = 2d \\ \mathrm{O}: &\quad 2b = 2c + d \end{aligned}$$

Set $a = 1$: $c = 2$, $d = 3$, $b = 7/2$. Multiply by 2:

$$2C_2H_6 + 7O_2 \to 4CO_2 + 6H_2O.$$



$CH_4 + 2O_2 \to CO_2 + 2H_2O$

Exercises

Practice 18.1. Balance (a) $\mathrm{Fe} + \mathrm{O_2} \to \mathrm{Fe_2O_3}$; (b) $\mathrm{C_3H_8} + \mathrm{O_2} \to \mathrm{CO_2} + \mathrm{H_2O}$.

Exercise 18.2. Write the stoichiometric matrix N for the reaction $\mathrm{CH_4} + 2\mathrm{O_2} \to \mathrm{CO_2} + 2\mathrm{H_2O}$ and verify $N\gamma = \mathbf{0}$.

Challenge 18.3. Balance $\mathrm{KMnO_4} + \mathrm{HCl} \to \mathrm{KCl} + \mathrm{MnCl_2} + \mathrm{H_2O} + \mathrm{Cl_2}$ by writing and solving the linear system. This is a non-trivial redox reaction with 7 unknowns.

Chapter 19

Moles and Quantitative Chemistry

19.1 Stoichiometric Calculations

Worked Example Mass of water produced

From $2H_2 + O_2 \rightarrow 2H_2O$, how many grams of water form from 4 g of H_2 (excess O_2)?

$$n(H_2) = \frac{4\ \text{g}}{2.016\,\text{g/mol}} \approx 1.984\,\text{mol}$$
$$n(H_2O) = 1.984\,\text{mol} \times \frac{2\ \text{mol}\ H_2O}{2\ \text{mol}\ H_2} = 1.984\,\text{mol}$$
$$m(H_2O) = 1.984 \times 18.016 \approx 35.7\ \text{g}$$

Exercises

Practice 19.1. From $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$, how many grams of CO_2 are produced by burning 11 g of propane ($M = 44.1\,\text{g/mol}$)?

Chapter 20

Reaction Ratios and Limiting Reagents

20.1 The Limiting Reagent

Definition Limiting Reagent

The **limiting reagent** is the reactant consumed first, setting the maximum possible yield. All other reactants are **excess reagents**.

Worked Example Juarrero: constraint propagation in a reaction

Consider $N_2 + 3H_2 \rightarrow 2NH_3$. Begin with 2 mol N_2 and 5 mol H_2.
N_2 alone could produce 4 mol NH_3; H_2 alone could produce $5 \times \frac{2}{3} \approx 3.33$ mol NH_3.
H_2 is limiting. Yield $= 3.33$ mol NH_3.
Juarrero [5] calls this *constraint cascade*: the scarcest resource propagates its limitation through the entire network.

20.2 Percent Yield

$$\%\text{yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%.$$

Exercises

Exercise 20.1. How many moles of excess N_2 remain after the reaction in the example above?

Challenge 20.2. Model a three-step reaction cascade $A + B \rightarrow C$, $C + D \rightarrow E$, $E + F \rightarrow G$. Starting with 4 mol each of A, B, D, F and none of the intermediates, determine the theoretical yield of G. Identify which reagent is limiting at each stage.

Chapter 21

Reaction Networks and Conservation Laws

21.1 Dynamic Stoichiometry

In a reaction network, concentrations $\mathbf{c}$ evolve by

$$\frac{d\mathbf{c}}{dt} = S\,\mathbf{r}(\mathbf{c}),$$

where S is the stoichiometric matrix and $\mathbf{r}(\mathbf{c})$ is the vector of reaction rates. Conservation is encoded in $\ker(S)$.

Worked Example Continuity in disguise

The equation above is the discrete analogue of the field-theoretic continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0,$$

describing conservation of density ρ in a flowing medium with velocity $\mathbf{v}$. Algebra, stoichiometry, and field theory all encode the same structural idea: *what flows in must flow out.*

Deeper Structure

Appendix E derives the RSVP field equations as the continuum limit of this conservation structure.

Part V

Integration: Mathematics of Structured Systems

Chapter 22

Constraint Networks

22.1 Networks as Mathematical Objects

A **network** (or graph) consists of **nodes** (vertices) and **edges** (links). When nodes represent quantities and edges represent constraints, the network encodes the structure of a system.

Worked Example Social constraint network after Arendt

Arendt's analysis of political space can be formalised as a constraint network in which nodes are actors and directed edges are communicative acts [1]. The system is stable when the network satisfies certain consistency conditions — analogous to the solvability of a linear system.

Worked Example Ecological resource balance after Le Guin

The allocation system $F = 2L$, $E = F + L$ from Chapter 4 is a two-node constraint network. Le Guin's utopian societies must maintain exactly such networks at equilibrium [3].

Chapter 23

Cycles, Flows, and Equilibrium

23.1 Three Kinds of Balance

All the subjects in this book converge on three structural ideas:

Type	Domain	Equation
Algebraic balance	variable equations	$ax = b$
Spatial balance	Pythagorean geometry	$a^2 + b^2 = c^2$
Rotational balance	unit circle	$\sin^2 + \cos^2 = 1$
Material balance	stoichiometry	$N\gamma = \mathbf{0}$
Dynamic balance	field continuity	$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$

23.2 Equilibrium as Fixed Point

Definition Equilibrium

A system state $\mathbf{x}^*$ is an **equilibrium** if applying the system's dynamics to $\mathbf{x}^*$ returns $\mathbf{x}^*$: $T(\mathbf{x}^*) = \mathbf{x}^*$.

Chapter 24

Scaling and Multilevel Structure

24.1 Patterns That Recur Across Scales

Theorem 24.1 (Self-Similarity Under Scaling)**.** *If a relationship $y = kx^p$ holds at one scale, and the same exponent p governs the relationship at larger or smaller scales, the pattern is* ***scale-invariant****.*

The inverse-square law $I = L/(4\pi r^2)$ is scale-invariant: it holds whether r is one metre or one light-year.

Worked Example Hazen: mineral evolution

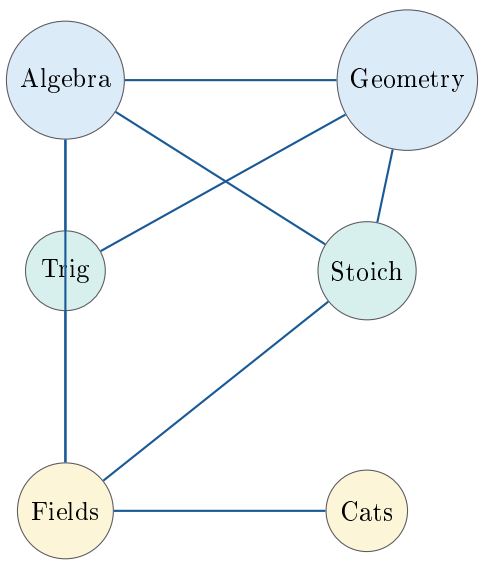
Hazen et al. [4] trace how the diversity of mineral species on Earth grew in proportion to the complexity of planetary processes. At each scale — atomic bonding, crystal structure, tectonic activity, biological metabolism — the same constraint logic applies: new mineral species form precisely when new constraints (temperature, pressure, chemical environment) are satisfied.

Chapter 25

From Elementary Systems to Scientific Models

25.1 The Architecture of Science

The tools introduced in this book form the foundation of all quantitative science. Every advanced model is an elaboration of the same constraint structure encountered in Chapter 1.



Guiding Principle

Scientific understanding grows by identifying patterns of constraint and transformation across domains. Transformation reveals structure; invariance reveals law.

Appendix A

Algebra as Constraint Before Content

A.1 The General Constraint System

Let $x_1, \ldots, x_n \in \mathbb{R}$ be system variables. A **constraint system** is a collection of equations

$$F_\alpha(x_1, \ldots, x_n) = 0, \quad \alpha = 1, \ldots, m.$$

The solution locus is $\mathcal{C} = \{x \in \mathbb{R}^n \mid F_\alpha(x) = 0 \; \forall \alpha\}$.

For linear constraints, $F_\alpha(x) = A_\alpha \cdot x - b_\alpha$, so the system becomes $A\mathbf{x} = \mathbf{b}$ with $A \in \mathbb{R}^{m \times n}$.

A.2 Admissibility and Relaxation

When $m > n$ the system is overdetermined and may be inconsistent. Define the **residual functional**

$$\mathcal{R}(\mathbf{x}) = \sum_{\alpha=1}^{m} |F_\alpha(\mathbf{x})|^2.$$

Approximate solutions minimise $\mathcal{R}$. This is the **least-squares principle**, which is also the precursor to entropic relaxation in the RSVP framework (Appendix E).

Proposition A.1 (Least Squares)**.** *The minimiser of* $\mathcal{R}(\mathbf{x}) = \|A\mathbf{x} - \mathbf{b}\|^2$ *satisfies the* ***normal equations*** $A^\top A\mathbf{x} = A^\top \mathbf{b}$.

A.3 Degrees of Freedom

When $m < n$, the system is underdetermined and the solution locus is a subspace of dimension $n - \mathrm{rank}(A)$. These degrees of freedom represent the system's capacity to vary while remaining consistent — a structural feature central to RSVP field configurations.

Appendix B

Geometry as Structured Invariance

B.1 Euclidean Space and the Inner Product

On $\mathbb{R}^n$, the standard inner product is $\langle x, y\rangle = \sum_{i=1}^n x_i y_i$, with induced norm $\|x\| = \sqrt{\langle x, x\rangle}$.

The distance formula and Pythagorean theorem both follow: $\|x - y\|^2 = \|x\|^2 - 2\langle x, y\rangle + \|y\|^2$.

B.2 Configuration Spaces

The RSVP theory works not over finite-dimensional $\mathbb{R}^n$ but over an infinite-dimensional configuration space:

$$\mathcal{M} = \left\{(\Phi, \mathbf{v}, S) \mid \Phi : \Omega \to \mathbb{R},\ \mathbf{v} : \Omega \to \mathbb{R}^3,\ S : \Omega \to \mathbb{R}\right\}.$$

An L^2-type inner product may be defined as

$$\langle X_1, X_2\rangle = \int_\Omega (\Phi_1\Phi_2 + \mathbf{v}_1 \cdot \mathbf{v}_2 + S_1 S_2)\, dx.$$

The Euclidean geometry of finite dimensions extends naturally to this infinite-dimensional setting.

Appendix C

Trigonometry as Oscillation, Rotation, and Recurrence

C.1 The Rotation Group

Define $R(\theta) \in \mathbb{R}^{2\times 2}$ as in Chapter 9. These matrices form the group SO(2) under composition, with $R(\theta_1)R(\theta_2) = R(\theta_1 + \theta_2)$.

The Pythagorean identity $\sin^2\theta + \cos^2\theta = 1$ is precisely the condition $\det R(\theta) = 1$.

C.2 Harmonic Analysis

The differential equation $\ddot{x} + \omega^2 x = 0$ governs simple harmonic motion. Its complex solutions are $e^{\pm i\omega t}$, where $e^{i\omega t} = \cos(\omega t) + i\sin(\omega t)$ (Euler's formula).

Linearising the RSVP system around a uniform background state yields modes of the form $e^{i(\mathbf{k}\cdot\mathbf{x} - \omega t)}$ — the same trigonometric recurrence, now in three spatial dimensions.

Appendix D

Stoichiometry as Conservation Algebra

D.1 The Stoichiometric Null Space

For a reaction network with stoichiometric matrix $S \in \mathbb{Z}^{e \times r}$ (elements $\times$ reactions), conservation is the condition $S\gamma = \mathbf{0}$. The admissible coefficient vectors form the **null space** $\ker(S)$.

D.2 Dynamic Stoichiometry

Concentrations $\mathbf{c}(t) \in \mathbb{R}^r$ evolve by

$$\dot{\mathbf{c}} = S\,\mathbf{r}(\mathbf{c}),$$

the *Chemical Master Equation* in its deterministic (mass-action) limit.

The connection to field theory: replacing $\mathbf{c}$ by a density field $\rho(x,t)$ and the finite sum by a spatial integral yields the continuity equation $\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$.

Appendix E

RSVP Field Theory: Foundational Derivation

E.1 The Fundamental Fields

Let $\Omega \subset \mathbb{R}^3$ be a spatial domain. The RSVP theory posits three coupled fields:

$$\Phi(x,t) \in \mathbb{R}, \quad \mathbf{v}(x,t) \in \mathbb{R}^3, \quad S(x,t) \in \mathbb{R},$$

representing plenum density, vector flow, and entropy density respectively.

E.2 Governing Equations

$$\partial_t \Phi + \nabla \cdot (\Phi \mathbf{v}) = D_\Phi \Delta \Phi + F_\Phi(\Phi, \mathbf{v}, S), \tag{A}$$
$$\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla)\mathbf{v} = -\nabla P(\Phi, S) + D_v \Delta \mathbf{v} + F_v(\Phi, \mathbf{v}, S), \tag{B}$$
$$\partial_t S + \nabla \cdot (S\mathbf{v}) = D_S \Delta S + \sigma(\Phi, \mathbf{v}, S). \tag{C}$$

Each left-hand side is an advection–transport operator; the diffusion terms smooth spatial gradients; the source terms encode nonlinear coupling.

E.3 Lyapunov Functional and Entropic Descent

Define

$$\mathcal{F}[\Phi, \mathbf{v}, S] = \int_\Omega \left[\tfrac{1}{2}|\mathbf{v}|^2 + U(\Phi, S) + \tfrac{\kappa_\Phi}{2}|\nabla \Phi|^2 + \tfrac{\kappa_S}{2}|\nabla S|^2\right] dx.$$

Dissipative dynamics descend on $\mathcal{F}$: $\partial_t X = -\frac{\delta \mathcal{F}}{\delta X}$. This makes precise the intuition that "space falls outward" toward lower free energy — the entropic smoothing picture underlying RSVP cosmology.

E.4 Linear Perturbation Theory

Expand around a homogeneous state $\Phi = \Phi_0 + \delta\Phi$, $\mathbf{v} = \delta\mathbf{v}$, $S = S_0 + \delta S$. The linearised system admits Fourier modes $\delta X \sim e^{i(\mathbf{k}\cdot x - \omega t)}$, whose dispersion relation $\omega(\mathbf{k})$ determines stability and propagation speed.

Appendix F

Entropic Redshift and Non-Expansion Cosmology

F.1 Redshift Without Metric Expansion

Let $\mathcal{E}(x,t)$ be an entropic attenuation field. Photon frequency along ray $x(\lambda)$ evolves as

$$\frac{d\nu}{d\lambda} = -\alpha\,\mathcal{E}(x(\lambda),t)\,\nu,$$

with solution

$$\nu(\lambda) = \nu_0 \exp\left(-\alpha \int_0^\lambda \mathcal{E}\,d\lambda'\right).$$

The redshift is $1+z = \exp\left(\alpha \int_0^\lambda \mathcal{E}\,d\lambda'\right)$. For weak, homogeneous $\mathcal{E}$:

$$z \approx \alpha\mathcal{E}\cdot\lambda,$$

recovering the observed linear distance–redshift law — without assuming metric expansion of space.

Appendix G

TARTAN: Recursive Tiling and Annotated Noise

G.1 Multiscale Partition

Let Ω admit a nested sequence of partitions $\mathcal{T}_0 \supseteq \mathcal{T}_1 \supseteq \cdots$ with refinement maps $\pi_k : \mathcal{T}_{k+1} \to \mathcal{T}_k$.

G.2 Annotated Field Updates

Each tile $T \in \mathcal{T}_k$ carries state $X_k(T) = (\Phi_k, \mathbf{v}_k, S_k, \eta_k)$, where η_k is a semantic / trajectory annotation.

Recursive update:

$$X_{k+1}(T') = F_k\big(X_k(\pi_k(T')),\, \xi_k(T')\big)\,,$$

where noise decomposes as $\xi = \xi_{\text{white}} + \xi_{\text{structured}}$, the structured part carrying provenance metadata.

History dependence is encoded by a trajectory buffer:

$$\mathcal{B}(T,t) = \{X(T, t-\tau_j)\}_{j=1}^{L}, \qquad X(T, t+\Delta t) = G(X(T,t), \mathcal{B}(T,t), \xi(T,t))\,.$$

Appendix H

Simulated Agency as Sparse Projection

H.1 The Projection Operator

Let $\mathcal{W}$ be the world state space and $\mathcal{M}$ the internal model space. An agent is characterised by a projection $P : \mathcal{W} \to \mathcal{M}$, $m = P(w)$.

Optimal projection minimises

$$m^* = \arg\min_m \big[L_{\text{task}}(m; w) + \beta\, C(m)\big],$$

trading off task loss against representational complexity.

H.2 The Agency Loop

$$w_{t+1} = F(w_t, a_t), \qquad m_t = P(w_t), \qquad a_t = \pi(m_t).$$

This recursive loop connects world, internal model, and action. It is a dynamical system on $\mathcal{W} \times \mathcal{M}$. Equilibria of the loop correspond to stable behaviours; periodic orbits correspond to habitual cycles. The geometric analysis of Chapter 9 applies directly.

Appendix I

Spherepop and Event-History Formalism

I.1 Events as Primitives

Let E be a set of events with causal order $\preceq$: $e_i \preceq e_j$ means e_i is causally prior to e_j.

A **state** is derived: $\sigma_t = \Sigma(E_{\leq t})$, where Σ is a summarisation operator.

Proposition I.1. *Ordinary state-update equations $x_{t+1} = f(x_t)$ arise as compressed approximations to richer event-structured evolution in which $E_{\leq t}$ is the fundamental datum.*

I.2 Admissible Futures

$$\mathrm{Poss}(E_{\leq t}) = \{e \in E \mid E_{\leq t} \cup \{e\} \text{ is causally consistent}\}.$$

This turns history into a *generator* of possible futures. The "append-only algebra" of Spherepop is precisely the monotone extension $E_{\leq t} \mapsto E_{\leq t} \cup \{e\}$ for $e \in \mathrm{Poss}(E_{\leq t})$.

Appendix J

Category-Theoretic Semantics of Constraint Systems

J.1 Categories

A **category** $\mathcal{C}$ consists of:

- objects $\mathrm{Ob}(\mathcal{C})$;
- for each pair A, B, a set $\mathrm{Hom}(A, B)$ of morphisms;
- an associative composition law $\circ$ and identity morphisms id_A.

Examples: Set (sets, functions); $\mathbf{Vect}_k$ (vector spaces, linear maps); **RxnNet** (species configurations, stoichiometric maps).

J.2 Symmetric Monoidal Structure

A **symmetric monoidal category** $(\mathcal{C}, \otimes, I)$ supports parallel composition. Semantic modules are objects; consistency-preserving transformations are morphisms; simultaneous operation is $\otimes$.

A merge operation, when it exists, is a **colimit**: the universal object receiving compatible maps from all merged components.

Appendix K

Sheaf-Theoretic Gluing and Local-to-Global Consistency

K.1 Presheaves and Sheaves

Let X be a topological space (the context space). A **presheaf** $\mathcal{F}$ assigns data $\mathcal{F}(U)$ to each open set U, with restriction maps $\rho_{UV} : \mathcal{F}(U) \to \mathcal{F}(V)$ for $V \subseteq U$.

$\mathcal{F}$ is a **sheaf** if: whenever sections $\{s_i \in \mathcal{F}(U_i)\}$ satisfy $\rho_{U_i,U_i\cap U_j}(s_i) = \rho_{U_j,U_i\cap U_j}(s_j)$ on all overlaps, there exists a unique global section $s \in \mathcal{F}(\bigcup U_i)$ restricting to each s_i.

Guiding Principle

Sheaf coherence is the precise mathematical form of local-to-global consistency. Every chapter of this book has been practicing local-to-global reasoning in disguise: from triangle ratios to stellar spectra, from stoichiometric balance to network equilibrium.

Appendix L

Variational Principles and Entropic Descent

L.1 The Euler-Lagrange Equation

Let $\mathcal{J}[u] = \int_\Omega \mathcal{L}(u, \nabla u, x)\, dx$. Critical points satisfy the **Euler-Lagrange equation**:

$$\frac{\partial \mathcal{L}}{\partial u} - \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial (\nabla u)} \right) = 0.$$

L.2 Gradient Flow

Dissipative dynamics minimising $\mathcal{J}$ satisfy:

$$\partial_t u = -\frac{\delta \mathcal{J}}{\delta u}.$$

This single equation subsumes: algebraic residual minimisation ($\mathcal{J} = \|A\mathbf{x} - \mathbf{b}\|^2$), geometric shortest paths, chemical equilibration, and entropic smoothing in RSVP — all are gradient flows on functionals appropriate to their domain.

Appendix M

Worked Derivations: Bridges Between the Introductory Subjects and the Theory

M.1 From Linear Algebra to Stoichiometry to Field Theory

Step 1. Balancing $2H_2 + O_2 \to 2H_2O$ is solving $S\gamma = \mathbf{0}$ with $S = \begin{pmatrix} 2 & 0 & 2 \\ 0 & 2 & 1 \end{pmatrix}$.

Step 2. Dynamic generalisation: $\dot{\mathbf{c}} = S\mathbf{r}(\mathbf{c})$.

Step 3. Continuum limit: replace $c_i(t)$ by $\rho(x,t)$ and the discrete sum by a spatial divergence, yielding $\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$.

Atom counting and field theory are the same idea at different scales.

M.2 From Distance Formula to L^2 Metric

Step 1. Euclidean distance $d(P,Q) = \|P - Q\|$ in $\mathbb{R}^n$.

Step 2. Extend to function space: $d(f,g) = \|f - g\|_{L^2} = \left(\int_\Omega |f-g|^2 dx\right)^{1/2}$.

Step 3. RSVP uses exactly this L^2 metric on field configurations to define entropic distance between system states.

M.3 From Trigonometric Oscillation to RSVP Eigenmodes

Step 1. $x(t) = A\cos(\omega t)$ solves $\ddot{x} + \omega^2 x = 0$.

Step 2. Linearise RSVP around uniform background. The perturbation $\delta\Phi$ satisfies $\partial_t^2(\delta\Phi) + \omega^2(\mathbf{k})\delta\Phi = 0$ for each Fourier mode $\mathbf{k}$.

Step 3. Each mode oscillates sinusoidally; the dispersion relation $\omega(\mathbf{k})$ encodes how the RSVP field propagates information.

Glossary

Amplitude
maximum displacement of a periodic function from its midline.

Avogadro's number
$N_A = 6.022 \times 10^{23}$; particles per mole.

Constraint
a condition restricting admissible values of variables.

Conservation law
a relation asserting that a quantity is preserved under transformation.

Dimensional analysis
tracking units through calculations.

Equilibrium
a fixed point of a dynamical system.

Event-history
a causally ordered sequence of occurrences; primitive in the Spherepop formalism.

Function
a rule assigning exactly one output to each input.

Gradient flow
dynamics $\partial_t u = -\delta\mathcal{J}/\delta u$ descending a functional.

Identity (trig.)
an equation true for all valid angles.

Invariant
a quantity unchanged by a transformation.

Isometry
a transformation preserving distances.

Limiting reagent
the reactant consumed first in a reaction.

Linear system
a system of equations of the form $A\mathbf{x} = \mathbf{b}$.

Molar mass
mass per mole of a substance (g/mol).

Mole
SI unit of chemical amount.

Null space
$\ker(A) = \{\mathbf{x} \mid A\mathbf{x} = \mathbf{0}\}$.

Period
smallest positive interval of repetition of a periodic function.

Pythagorean theorem
$a^2 + b^2 = c^2$ for a right triangle.

Radian
angle subtended by arc of length equal to radius.

RSVP
Relativistic Scalar-Vector Plenum field theory.

Sheaf
a presheaf satisfying the gluing axiom for local data.

Similarity
geometric relation in which figures differ only by scale.

Slope
rate of change of a linear function.

Stoichiometric matrix
element-species incidence matrix S encoding conservation.

TARTAN
Recursive Tiling and Annotated Noise simulation framework.

Unit circle
circle of radius 1; foundation of trigonometric definitions.

Variable
a letter representing an unknown or varying quantity.

Variational functional
a function on function spaces, $\mathcal{J}[u]$.

Selected Answers

Chapter 2 (Equations and Balance). Practice 2.1: (a) $x = 5$; (b) $x = 15$; (c) $x = 5$; (d) $x = 7$.

Chapter 7 (Pythagorean Theorem). Practice 7.1: diagonal $= \sqrt{49 + 576} = 25$ cm.

Chapter 10 (Atoms and Moles). Practice 10.1: (a) 58.44; (b) 44.01; (c) 180.16 g/mol.

Chapter 17 (Balancing). Practice 17.1: (a) $4\text{Fe} + 3\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3$; (b) $\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$.

Chapter 18 (Moles). Practice 18.1: ≈ 33.6 g CO_2.

Abbreviated Periodic Table

Symbol	**Name**	Z	M **(g/mol)**
H	Hydrogen	1	1.008
C	Carbon	6	12.011
N	Nitrogen	7	14.007
O	Oxygen	8	15.999
Na	Sodium	11	22.990
Mg	Magnesium	12	24.305
S	Sulfur	16	32.060
Cl	Chlorine	17	35.453
K	Potassium	19	39.098
Ca	Calcium	20	40.078
Fe	Iron	26	55.845
Cu	Copper	29	63.546

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